

Literature Review Detailed: Key Results and Intuition

Draft notes

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This note records, paper by paper, the key results and the intuition behind them. It is more detailed than the positioning note (`literature_view`).

1 Public finance and deadweight loss

1.1 The Measurement of Waste (Harberger, 1964)

Harberger (1964) develops a linearized framework for measuring aggregate efficiency losses from taxes and related price distortions. The paper emphasizes six results.

- **Measuring deadweight loss (ΔW) as a policy tool**

- **Key result:** Deadweight loss can be calculated with a linearized aggregate framework that measures efficiency changes across distorted markets:

$$\Delta W = \frac{1}{2} \sum_i \sum_j R_{ij} T_i T_j.$$

- **Intuition:** Rather than only asking whether an allocation is Pareto optimal, policy-makers need a quantitative measure of efficiency loss to weigh tax reforms, tariffs, and other real-world trade-offs.

- **Neutrality of uniform value-added taxes**

- **Key result:** An equal percentage tax on value added across all productive activities yields zero net efficiency loss:

$$\sum_j v_j R_{ij} = 0 \implies \Delta W = 0 \quad \text{when } t_i = t \text{ for all } i.$$

- **Intuition:** Uniform value-added taxation leaves relative price ratios across activities unchanged, so it does not induce distortionary factor substitution.

- **Equivalence of product taxes and value-added taxes**

- **Key result:** Product taxes can be translated into equivalent taxes on value added when intermediate inputs enter output in fixed proportions.
- **Intuition:** Taxing final output creates the same behavioral incentives and equilibrium as taxing all factor inputs (labor, capital, materials) at the same effective rate.

- **Cross-tax distortions and welfare-improving tax increases**

- **Key result:** Introducing or raising a tax on a good can raise total social welfare if that good is a substitute for another already heavily taxed good:

$$\frac{\partial(\Delta W)}{\partial T_k} > 0 \quad \text{when } T_j > 0 \text{ and } R_{kj} > 0.$$

- **Intuition:** If Good A is heavily taxed, consumers substitute toward Good B . Taxing Good B shifts demand back toward Good A , reducing the preexisting distortion in Sector A by enough to offset the new distortion in Sector B .

- **Distortion of the labor–leisure choice**

- **Key result:** A uniform tax on all market labor is non-neutral unless leisure (non-market activity) can also be taxed:

$$\sum_{j=1}^{n+1} M_{ji} = 0.$$

- **Intuition:** Leisure is hard to tax. Taxing all market labor therefore induces substitution into untaxed leisure and creates an efficiency loss in labor supply.

- **Capital–labor cross-factor interaction effects**

- **Key result:** When taxes distort both capital (B_i) and labor (E_i), cross-factor interaction terms ($H_{ij} = \partial K_i / \partial E_j$) must be included to measure total efficiency loss:

$$\Delta W = \frac{1}{2} \sum_i \sum_j M_{ij} E_i E_j + \frac{1}{2} \sum_i \sum_j G_{ij} B_i B_j + \sum_i \sum_j H_{ij} B_i E_j.$$

- **Intuition:** A tax on capital in one industry reallocates both capital and labor across sectors. Ignoring how labor responds to capital taxation (and vice versa) misstates total efficiency loss.

1.2 Salience and Taxation: Theory and Evidence (Chetty, Looney, & Kroft, 2009)

Chetty et al. (2009) show that consumer responses to taxation depend on whether the tax is salient at the point of choice. The paper combines a field experiment with evidence from alcohol taxation to argue that standard tax theory misses an important channel when buyers do not fully perceive non-salient taxes.

- **Underreaction to non-salient taxes**

- **Key result:** Consumers consistently underreact to taxes that are not explicitly included in posted price tags.
- **Intuition:** Standard models assume buyers continuously compute the full final price. In practice, doing so requires active mental effort, so many consumers fail to fully incorporate taxes that are added only at checkout.

- **The salience parameter θ**

- **Key result:** The paper summarizes attention with a salience parameter $\theta \in [0, 1]$, so consumers behave as if they face the perceived price $\tilde{p} = p + \theta t$ rather than the true price $p + t$.
- **Intuition:** When $\theta = 1$, the model collapses to the standard full-optimization benchmark and consumers treat a tax increase like a base-price increase. When $\theta = 0$, consumers ignore the tax completely while shopping. The empirically relevant case is $0 < \theta < 1$, where consumers notice the tax but underreact to it.

- **Grocery store field experiment**

- **Key result:** Posting tax-inclusive shelf tags reduced demand by about 8 percent.
- **Intuition:** Consumers rely heavily on the visible shelf price as a decision shortcut. Making the tax visible at the moment of choice changes behavior because it converts a hidden cost into an immediately attended price signal.

- **Excise tax versus sales tax impact**

- **Key result:** State-level increases in alcohol excise taxes, which are embedded in posted prices, reduced beer consumption by an order of magnitude more than equivalent increases in general sales taxes, which are added at checkout.
- **Intuition:** Taxes that are already incorporated in shelf prices affect the price consumers actually see while shopping, whereas register-added taxes remain less salient until after most purchase decisions are made.

- **Economic incidence depends on statutory form**

- **Key result:** Economic incidence depends on the statutory form of the tax once salience matters, so excise taxes and register-added sales taxes need not place the same burden on consumers and producers.
- **Intuition:** When a tax is non-salient, consumers do not reduce demand as much as they would under full attention. Because quantity demanded remains artificially high, producers do not need to cut pre-tax prices as much to sustain sales, and inattentive consumers bear a larger share of the burden. This overturns the standard view that statutory form is irrelevant for incidence.

- **Incidence formula adjustment under inattention**

- **Key result:** In the standard model, the producer burden depends on relative elasticities, with Burden on producers $\propto \epsilon_D/(\epsilon_S + \epsilon_D)$. In the behavioral model, the demand response to a tax is attenuated to $\theta\epsilon_D$.
- **Intuition:** Because consumers react only to the salient fraction of the tax, demand falls less after a tax increase than a standard model predicts. Producers therefore do not need to lower pre-tax prices as much, so more of the tax is shifted onto consumers. In the limiting case $\theta = 0$, demand does not respond to the tax at all and producers can pass through the full tax without absorbing any of the burden through lower prices.

- **Decision miscalculation and overconsumption**

- **Key result:** Consumers choose based on a perceived price $\tilde{p} = p + \theta t$ with $\theta < 1$, rather than the true tax-inclusive price $p + t$, so non-salient taxes can generate excess burden even if observed demand barely moves.

- **Intuition:** Consumers buy extra units whose marginal benefit is below their true marginal cost because the tax is only partially internalized at the time of choice. Section V of the paper formalizes this point: decision errors themselves generate deadweight loss, even when observed quantity responses are very small or zero.

- **Deadweight-loss formula adjustment**

- **Key result:** Welfare loss from demand distortion shrinks with tax inattention, so a useful generalized Harberger benchmark is $DWL_{\text{behavioral}} \approx \frac{1}{2}pQ\epsilon_D\theta t^2$. But total consumer welfare still falls because inattentive consumers make costly decision errors even when observed demand changes very little ($\Delta Q \approx 0$).
- **Intuition:** Standard tax analysis measures harm mainly by how many purchases are canceled. When taxes are hidden, a small ΔQ does not mean the tax is harmless; it means consumers are being induced to buy goods whose true cost exceeds their value. Welfare losses therefore arise through optimization mistakes and unexpected budget squeezes, not only through the usual demand-drop channel.

- **Unexpected budget squeezes and suboptimal cutbacks**

- **Key result:** Non-salient taxes can worsen welfare by causing consumers to violate their intended budget allocation while browsing and then forcing unplanned adjustments at checkout.
- **Intuition:** Because the omitted tax is not reflected in the posted price, consumers do not initially choose bundles that satisfy their true budget constraint $Z = (p + t)x + y$. Once the tax is charged, they must cut spending on other goods unexpectedly, ending up with a worse overall consumption bundle than they would have chosen under full information.

References

- Raj Chetty, Adam Looney, and Kory Kroft. Salience and Taxation: Theory and Evidence. *American Economic Review*, 99(4):1145–1177, August 2009. ISSN 0002-8282. doi: 10.1257/aer.99.4.1145. URL <https://pubs.aeaweb.org/doi/10.1257/aer.99.4.1145>.
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